

University of
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Visualising Emotion: A Similarity-Based Analysis of Emotional Fidelity in Text-to-Image Generation

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Declaration

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Abstract

Glossary

Record One prompt–drawing pair from the corpus. Records are not linked to individuals; 1,025 were processed and 1,012 analysed.

Prompt reader The text model that reads the visitor's words: GoEmotions (RoBERTa fine-tuned on 28 labels), collapsed to seven families.

Describer The vision model that answers five factual questions about a drawing (BLIP-VQA). It is never asked for an emotion.

Image reader The text model that infers an emotion family from the describer's one-sentence description (BART-MNLI, zero-shot).

Family One of seven emotion families: joy, sadness, anger, fear, surprise, disgust, neutral (Ekman's six plus neutral).

Agreement The central quantity: how similar the prompt reader's reading of the words is to the image reader's reading of the drawing. It is a property of two machine readers, not a statement about what a person would see.

Valence sign + for joy, − for sadness, anger, fear and disgust, 0 for surprise and neutral. Valence-sign agreement is computed over records whose prompt family has a sign.

Family agreement The share of records whose top-1 families coincide.

Top-3 soft overlap The overlap of the two readers' top-3 distributions after each is normalised to sum to one.

Valence–arousal similarity One minus the normalised distance between the two readings' score-weighted centroids in valence–arousal space. Its floor over the family centroids is 0.412, and randomly paired records already average about 76.6%.

Random-pairing null The mean and spread of a measure when the image readings are re-paired with the prompts at random (300 re-pairings, seed 7). Every reported value is read against it.

z (observed − null mean) / null standard deviation. An ordering statistic here, not a precise probability: across seeds it moves by about one unit (+11.2 to +12.2 for the headline).

Degenerate null A null with zero spread, which arises when every prompt in a subset shares one label so permutation cannot change the score. No z is reported for it.

Prompt group / dedup_key All records whose prompt text (before redaction, case-folded) is identical. 763 groups; 249 records repeat an earlier prompt. Every resampling and permutation clusters on it.

Clustered bootstrap Resampling prompt groups, not records, with replacement (10,000 draws) to obtain 95% confidence intervals that respect repeated prompts.

Face-classified / scene-classified A record is face-classified when the describer reports at least one person *and* its expression answer contains a readable expression word; every other record is scene-classified. The split is the describer's, not a human's.

Emotional / neutral-labelled prompt A prompt is emotional when the prompt reader's top-1 family is not neutral (607); otherwise it is neutral-labelled (405). The label is the reader's judgement, never human-checked.

Specificity check The expectation that neutral-labelled prompts show no agreement above their null. Only the valence–arousal measure is informative there.

Copy rate Where the describer's expression answer names an emotion word, the share of records in which the image reader's top-1 family equals that word's family (99.8% on the strict definition).

Consistency For prompts drawn more than once, the mean share of drawings in a group that the image reader assigns the group's modal family, read against a permutation null.

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2.2 Foundations: emotion theory and taxonomy

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Chapter 3

Methodology

3.1 Overview and design principles

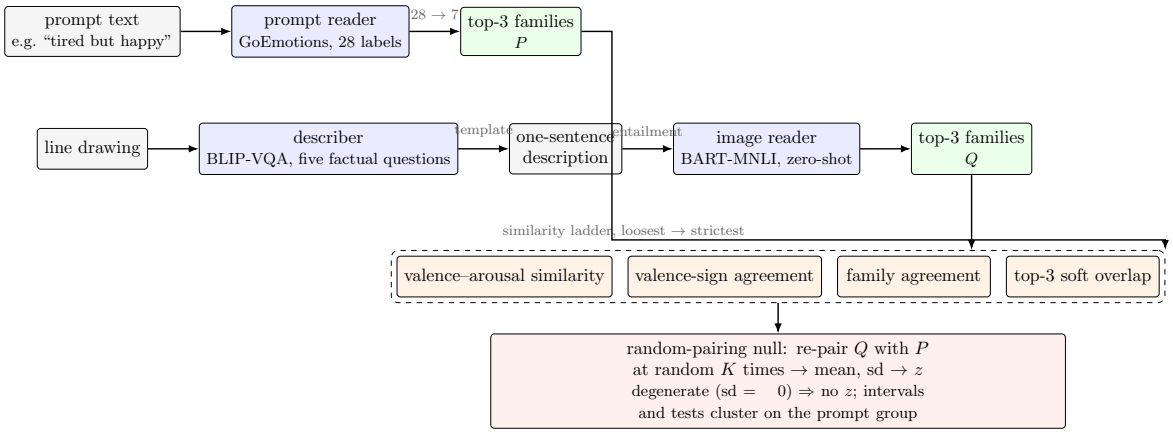


Figure 3.1: The describe-then-extract pipeline and the similarity ladder. The prompt reader (GoEmotions) and the image reader (BART-MNLI over a BLIP-VQA description) each return a top-3 distribution over seven families; four measures compare them, and each measure is read against the mean and spread it takes when the image readings are re-paired with the prompts at random.

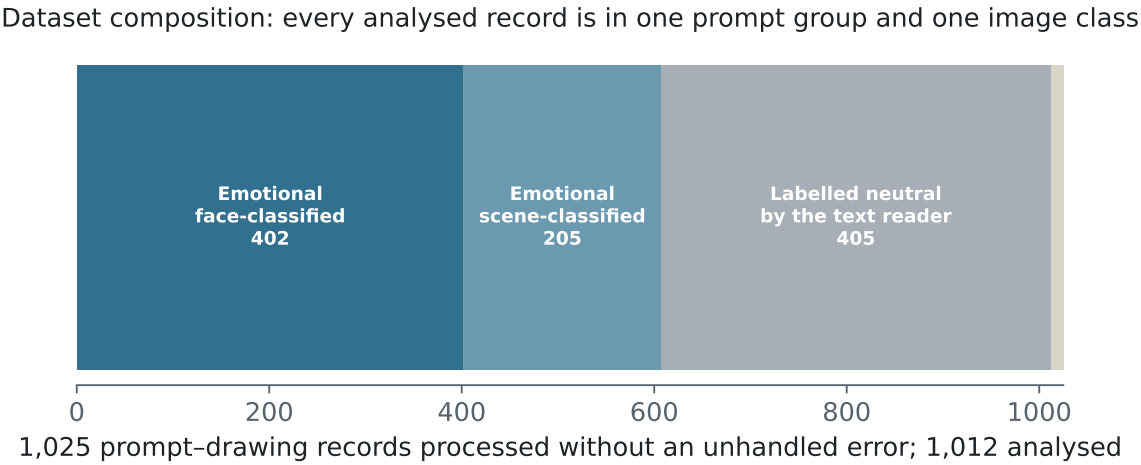


Figure 3.2: Composition of the corpus: 1,025 records processed, 13 pre-festival developer tests excluded, 1,012 analysed. Every analysed record belongs to exactly one prompt group (emotional / neutral-labelled) and one image class (face-classified / scene-classified); the groups are groupings, not exclusions.

3.2 The corpus

3.3 Data pre-processing

3.3.1 Loading and pre-test removal

3.3.2 Prompt groups

3.3.3 Name redaction

3.4 The pipeline

3.4.1 Prompt reader

3.4.2 Describer

3.4.3 Description template and image reader

3.4.4 Face and scene classes

3.5 The similarity ladder

Let each reader return a top-3 list $T = \{(\ell_i, s_i)\}_{i=1}^{3}$ of families $\ell_i \in \mathcal{F}$ with scores s_i, where $\mathcal{F} = \{\text{joy}, \text{sadness}, \text{anger}, \text{fear}, \text{surprise}, \text{disgust}, \text{neutral}\}$. Write P for the prompt reader's list and Q for the image reader's, and $\text{top}(T) = \ell_1$ for the top-1 family. Each family has a fixed centroid $c_\ell = (v_\ell, a_\ell)$ on the valence–arousal circumplex (Appendix B).

Valence–arousal similarity. The score-weighted centroid of a reading and the similarity between two readings are

$$\mathbf{v}(T) = \frac{\sum_i s_i\, c_{\ell_i}}{\sum_i s_i}, \qquad S_{\mathrm{VA}}(P,Q) = 1 - \frac{1}{\sqrt{2}} \left\| \left(\tfrac{v_P+1}{2} - \tfrac{v_Q+1}{2},\ a_P - a_Q \right) \right\|_2, \tag{3.1}$$

where $(v_P, a_P) = \mathbf{v}(P)$ and likewise for Q. Valence is rescaled to $[0,1]$ so both axes have unit range. Over the seven centroids the smallest attainable value is 0.412 (joy against sadness), and randomly paired records average about 76.6%; this floor is why the measure is never quoted without its null.

Top-3 soft overlap. Each side is first normalised to a distribution over its own three families, $\tilde{p}_\ell = s_\ell / \sum_j s_j$, so the prompt reader's multi-label scale and the image reader's softmax scale are comparable; the overlap is the sum of per-family minima,

$$S_{\mathrm{ovl}}(P,Q) = \sum_{\ell \in \mathcal{F}} \min\!\left(\tilde{p}_\ell,\ \tilde{q}_\ell\right) \in [0,1], \tag{3.2}$$

which equals one only when the two distributions coincide.

Family agreement. Over a set of n records,

$$A_{\mathrm{fam}} = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\big[\mathrm{top}(P_i) = \mathrm{top}(Q_i)\big]. \tag{3.3}$$

Valence-sign agreement. With $\sigma(\ell) \in \{+, -, 0\}$ the sign of a family ($+$ for joy; $-$ for sadness, anger, fear, disgust; 0 for surprise and neutral),

$$A_{\mathrm{sign}} = \frac{\sum_i \mathbb{1}\big[\sigma(\mathrm{top}\, P_i) \neq 0 \ \wedge\ \sigma(\mathrm{top}\, P_i) = \sigma(\mathrm{top}\, Q_i)\big]}{\sum_i \mathbb{1}\big[\sigma(\mathrm{top}\, P_i) \neq 0\big]}, \tag{3.4}$$

so records whose prompt family has no sign are excluded from the denominator; the denominator is reported with every value (for example 530 of the 607 emotional prompts).

3.6 The random-pairing null

For a subset of n records and a measure m, draw K random permutations π_k of the image readings and compute $m_k = m\big(\{(P_i, Q_{\pi_k(i)})\}_{i=1}^{n}\big)$. With $\hat{\mu} = \frac{1}{K}\sum_k m_k$ and $\hat{\sigma}$ the standard deviation of $\{m_k\}$,

$$z = \frac{m_{\mathrm{obs}} - \hat{\mu}}{\hat{\sigma}}, \qquad K = 300,\ \text{seed } 7. \tag{3.5}$$

A null is *degenerate* when $\hat{\sigma} = 0$: this happens for family agreement when every prompt in the subset shares one top-1 family, and for valence-sign agreement when every prompt shares one

sign, because permuting the image side then cannot change the score. The analysis detects this from the null itself and prints no z for such cells.

3.7 Inference

Clustered bootstrap confidence intervals. Let $\mathcal{G}$ be the set of prompt groups in a subset. For $b = 1, \ldots, B$ draw $|\mathcal{G}|$ groups with replacement, pool their records, and recompute the measure; the 95% interval is the 2.5th and 97.5th percentiles of the B values [22], with $B = 10,000$ and seed 7.

Clustered permutation test of the face/scene difference. Under the null that the image class is unrelated to agreement, the face/scene labels are exchangeable across prompt groups. The labels are permuted across groups P times and the difference in agreement recomputed; the two-sided p-value uses the estimator of [63],

$$p = \frac{k+1}{P+1}, \qquad k = \#\{\text{permuted differences at least as extreme as the observed}\}, \tag{3.6}$$

so the smallest reportable value with $P = 10,000$ is 10^{-4}.

Cohen's kappa. For comparability with other studies, family agreement is also reported as

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{3.7}$$

with p_o the observed top-1 agreement and p_e the agreement expected from the two readers' family marginals [18], interpreted with the bands of [40]; intervals by the clustered bootstrap above.

Multiple comparisons. The analysis prints 36 non-degenerate z-scores. Each is converted to a two-sided p by the normal approximation and the family is controlled with the Holm–Bonferroni step-down procedure [31]: order $p_{(1)} \leq \cdots \leq p_{(m)}$ and reject while $p_{(i)} \leq \alpha/(m - i + 1)$, $\alpha = 0.05$.

Paired comparison of two image readers. On the same records each reader either agrees with the prompt reader or does not. With b the records only the first reader agrees on and c only the second, McNemar's exact binomial test [52] on (b, c) treats records as independent and is therefore optimistic here; the clustered version flips the sign of each prompt group's summed difference as a block (20,000 draws) and the clustered bootstrap gives the interval on the difference in rates.

Wilson intervals. Proportions from the single-coder check use the Wilson score interval [80], which behaves at small n and near 0 and 1.

3.8 Analyses

Consistency across repeated prompts. For a prompt group g with $n_g \geq 2$ records and image top-1 families $\{\text{top}(Q_i)\}_{i \in g}$, the modal fraction is $c_g = \max_\ell n_{g,\ell}/n_g$, and the consistency is the mean $C = \frac{1}{|\mathcal{G}_{\geq 2}|} \sum_g c_g$ over the 87 groups. Because 43 groups are pairs, c_g cannot fall below 0.5 for them, so C is read against a permutation null that shuffles the image top-1 families across all 1,012 analysed records and re-cuts them into the same groups (10,000 draws). The count of fully consistent groups ($c_g = 1$) is treated the same way. This bounds the reproducibility of the whole prompt-to-reading chain; it does not isolate the generator, whose parameters were not recorded.

Copy rate. Where the describer's expression answer is a single word that names an emotion (strict: happy, sad, angry, ...; 418 records) or a facial action with an implied family (broad: smile, crying, ...; 997 records), the copy rate is the share of those records in which the image reader's top-1 family equals the family the word implies. It measures how much the "separate" extraction stage adds beyond the describer's own word.

3.9 The demonstrator

3.10 Reproducibility and testing

Chapter 4

Results

4.1 Overall agreement

Table 4.1: Agreement between the prompt reader and the image reader on the 1,012 analysed records, by prompt group and image class, against the random-pairing null (300 re-pairings, seed 7). CI: clustered bootstrap over prompt groups (10,000 draws). n/a: degenerate null (every prompt in the subset shares one label, so permutation cannot change the score).

Subset / measure	Observed	Null	z	95% CI
Emotional prompts (n=607; sign n=530)				
Valence–arousal similarity	83.8%	77.3%	+12.2	[82.0, 85.5]
Valence-sign agreement	64.0%	45.6%	+11.1	[58.6, 68.8]
Family agreement	53.0%	38.8%	+10.3	[47.3, 58.7]
Top-3 soft overlap	50.6%	39.4%	+11.6	[46.5, 54.7]
face-classified (n=402; sign n=360)				
Valence–arousal similarity	84.2%	76.4%	+11.2	[82.3, 86.1]
Valence-sign agreement	73.9%	53.5%	+10.4	[68.3, 79.2]
Family agreement	61.9%	45.8%	+10.3	[55.8, 68.0]
Top-3 soft overlap	53.4%	40.4%	+11.6	[49.2, 57.7]
scene-classified (n=205; sign n=170)				
Valence–arousal similarity	82.9%	79.1%	+6.0	[80.4, 85.2]
Valence-sign agreement	42.9%	29.6%	+5.8	[33.1, 51.1]
Family agreement	35.6%	26.2%	+4.4	[26.4, 44.1]
Top-3 soft overlap	44.9%	38.1%	+4.9	[38.7, 51.0]
Neutral-labelled prompts (n=405; sign n=0)				
Valence–arousal similarity	76.6%	76.6%	+0.3	[75.6, 77.6]
Valence-sign agreement	0.0%	0.0%	n/a	degenerate
Family agreement	28.6%	28.6%	n/a	degenerate
Top-3 soft overlap	31.6%	31.6%	+0.0	–

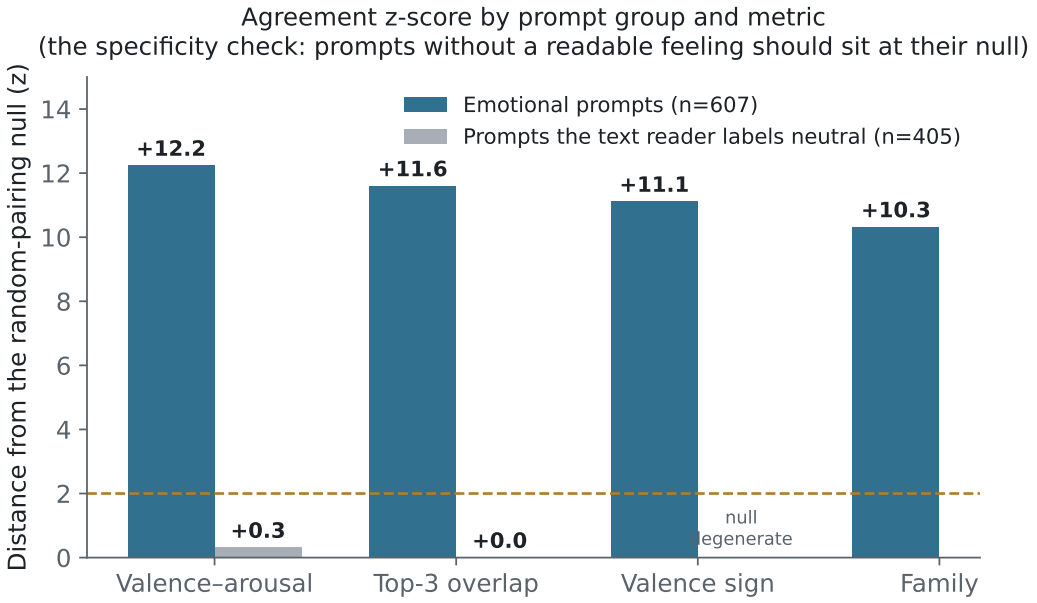


Figure 4.1: Distance from the random-pairing null (z) by prompt group and measure. For the neutral-labelled prompts the sign and family nulls are degenerate and no z is shown.

4.2 Face-classified versus scene-classified drawings

Table 4.2: Face-classified versus scene-classified drawings on the emotional prompts: agreement with 95% clustered-bootstrap CIs, the difference with its CI, and the clustered permutation p (10,000 draws; significance.py).

Measure	Face (n=402)	Scene (n=205)	Difference (pp)	Perm. p
Valence-sign agreement	73.9% [68.3, 79.2]	42.9% [33.1, 51.1]	+30.9 pp [+22.6, +40.6]	0.0001
Family agreement	61.9% [55.8, 68.0]	35.6% [26.4, 44.1]	+26.3 pp [+18.4, +35.3]	0.0001

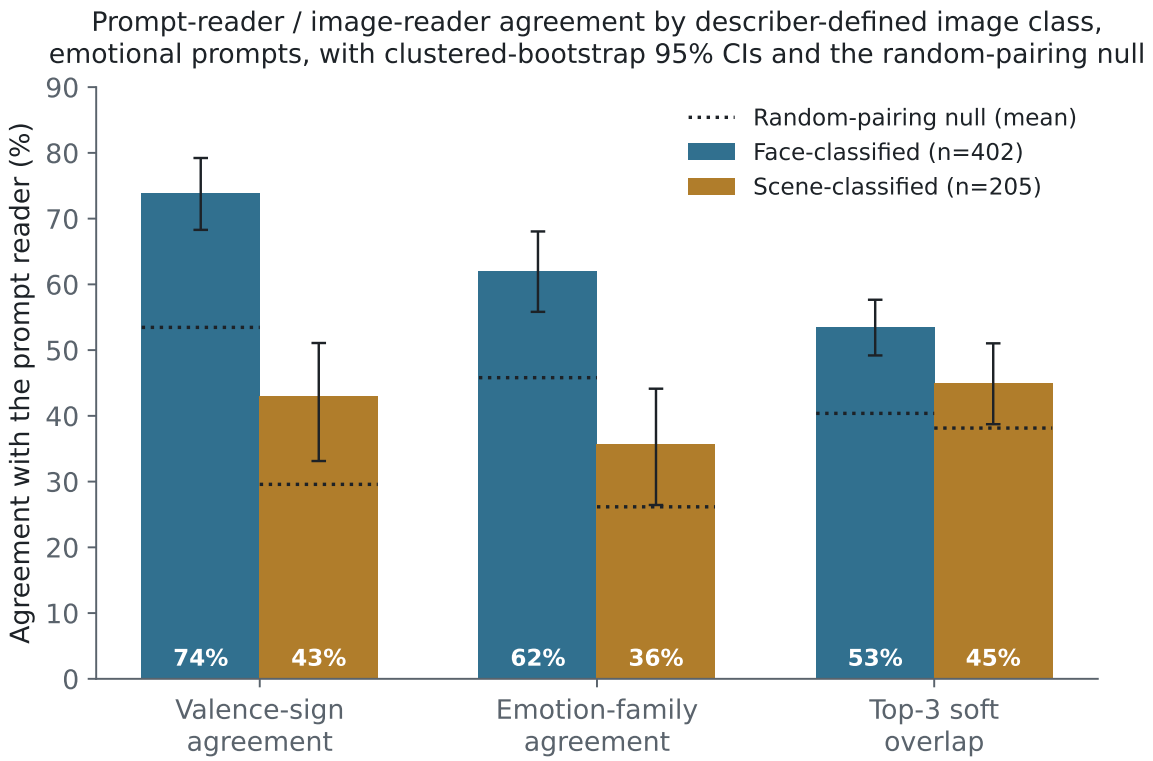


Figure 4.2: Agreement between the prompt reader and the image reader on the emotional prompts, by describer-defined image class, with clustered-bootstrap 95% CIs (error bars) and each cell's random-pairing null (dotted).

Table 4.3: The same analysis on the 763 distinct prompts only (first occurrence of each prompt text kept), a check against pseudo-replication from the 249 repeated prompts.

Subset / measure	Observed	Null	z	95% CI
Emotional prompts (n=452; sign n=394)				
Valence–arousal similarity	83.0%	76.5%	+10.6	[81.5, 84.4]
Valence-sign agreement	62.2%	44.4%	+9.9	[57.2, 67.0]
Family agreement	51.3%	37.2%	+9.3	[46.7, 56.0]
Top-3 soft overlap	48.9%	37.7%	+10.8	[45.9, 51.9]
face-classified (n=319; sign n=284)				
Valence–arousal similarity	83.5%	75.5%	+9.7	[81.7, 85.2]
Valence-sign agreement	71.8%	51.0%	+8.4	[66.4, 77.0]
Family agreement	59.9%	43.0%	+8.1	[54.2, 65.2]
Top-3 soft overlap	51.8%	38.4%	+9.7	[48.1, 55.4]
scene-classified (n=133; sign n=110)				
Valence–arousal similarity	81.8%	78.9%	+3.6	[79.5, 83.8]
Valence-sign agreement	37.3%	27.1%	+3.5	[28.1, 46.2]
Family agreement	30.8%	24.1%	+2.6	[23.3, 38.4]
Top-3 soft overlap	41.7%	36.7%	+3.1	[36.9, 46.5]
Neutral-labelled prompts (n=311; sign n=0)				
Valence–arousal similarity	76.6%	76.4%	+1.1	–
Valence-sign agreement	0.0%	0.0%	n/a	degenerate
Family agreement	27.7%	27.7%	n/a	degenerate
Top-3 soft overlap	31.4%	31.1%	+0.8	–

Table 4.4: Multiple-comparison control over the 36 non-degenerate z-scores the analysis prints (12 degenerate ones excluded). Two-sided p by the normal approximation; Holm–Bonferroni at family-wise 0.05: 26 survive, 10 do not.

Outcome	Tests
Do not survive	DISTINCT negative prompts overlap z+2.7; DISTINCT scene images family z+2.6; FULL positive prompts overlap z+2.3; FULL negative prompts va z+2.1; DISTINCT positive prompts overlap z+1.9; DISTINCT negative prompts va z+1.6; DISTINCT non-emotional va z+1.1; DISTINCT non-emotional overlap z+0.8; FULL non-emotional va z+0.3; FULL non-emotional overlap z+0.0
Survive	the remaining 26, including every headline (emotional, face and scene on all four measures; distinct-prompt emotional and face)

Table 4.5: The specificity check on the prompts the text reader labels neutral. Only valence–arousal similarity is informative there (sign and family are degenerate: every prompt shares the neutral label). Cleaned control = minus the 2 records reduced to '[name]' and the 68 containing one of 66 plain affect words.

Control set	n	VA observed	VA null	z (record null)	z (clustered null)
All neutral-labelled prompts	405	76.6%	76.5%	+0.5	-0.1
Cleaned control	335	76.6%	76.6%	+0.0	-0.6

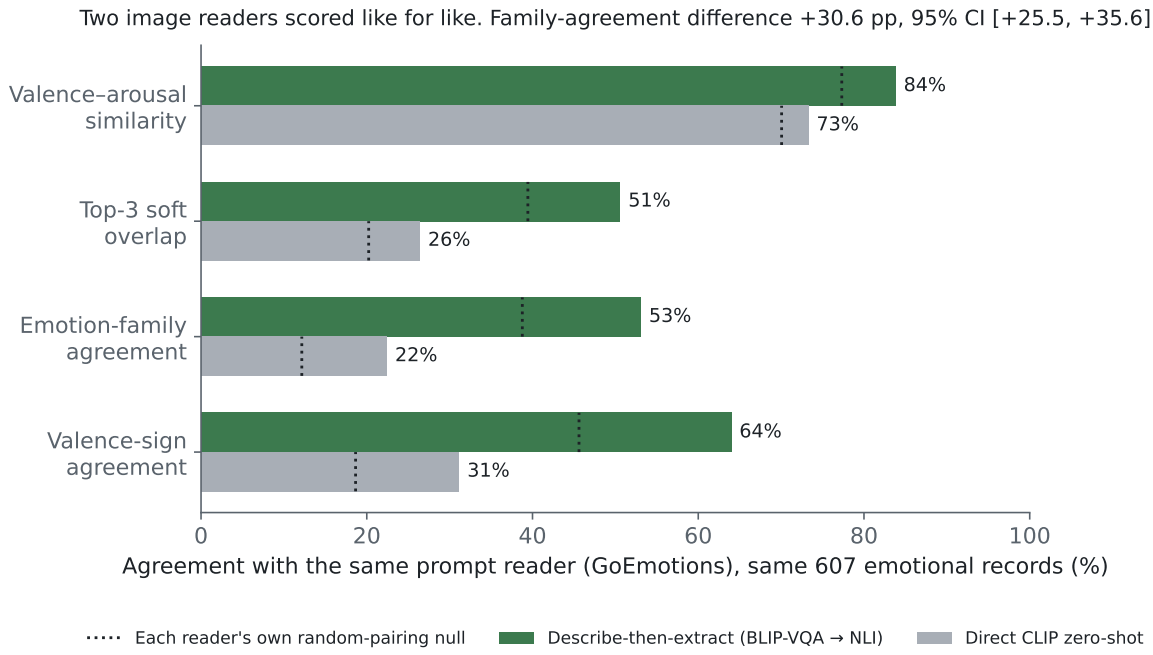


Figure 4.3: Two image readers scored against the same prompt reader on the same 607 emotional records, each against its own random-pairing null (dotted).

Table 4.6: Two image readers scored like for like against the same prompt reader (GoEmotions) on the same 607 emotional records with the same measures and the same 300 re-pairings. b = records only describe-then-extract agrees on; c = records only direct CLIP agrees on. The exact McNemar p treats records as independent and is optimistic (3.4×10^{-35} for family); the cluster sign-flip randomisation (20,000 draws) respects prompt groups.

Measure	Describe→extract	null	z	Direct CLIP	null	z
Valence–arousal similarity	83.8%	77.3%	+12.2	73.3%	70.1%	+10.2
Valence-sign agreement	64.0%	45.6%	+11.1	31.1%	18.6%	+9.9
Family agreement	53.0%	38.8%	+10.3	22.4%	12.2%	+11.1
Top-3 soft overlap	50.6%	39.4%	+11.6	26.4%	20.2%	+11.2
Paired family test	b=218, c=32			diff +30.6 pp	CI [+25.5, +35.6]	cluster $p < 0.001$
Paired valence-sign test	b=204, c=30			diff +32.8 pp	CI [+27.4, +38.2]	cluster $p < 0.001$

Table 4.7: The earlier (v1) comparison as published in the build log: prompt side read by BART-MNLI, the 608 records of the v1 lexicon rule, and a different overlap formula. Kept as a record; not comparable with Table 4.1.

Measure	Describe→extract	Direct CLIP
Family agreement	49.3%	25.3%
Top-3 soft overlap	48.6%	22.8%
Valence–arousal similarity	83.7%	74.2%

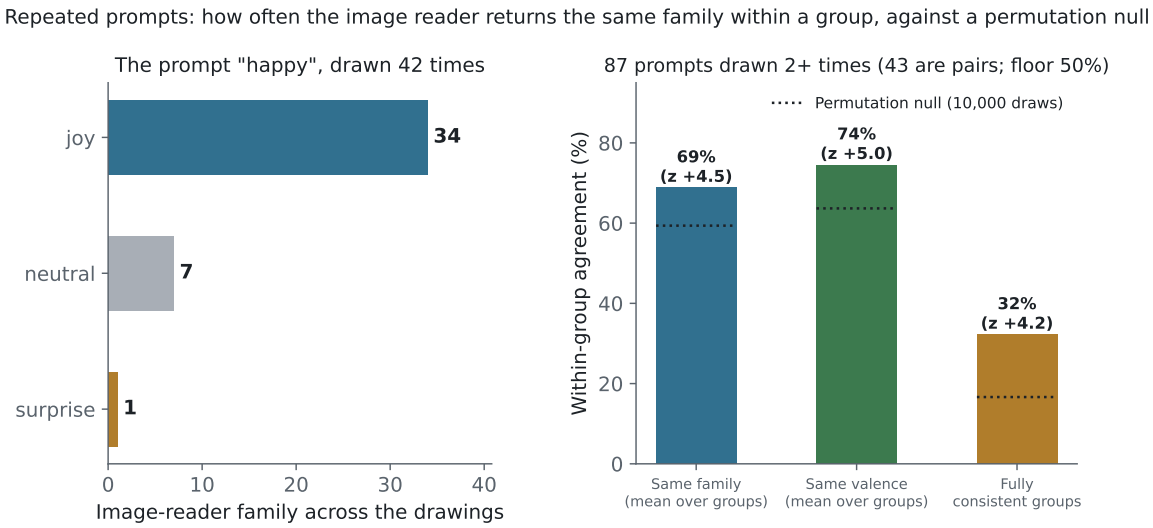


Figure 4.4: Repeated prompts: the image reader's top-1 family across the 42 drawings of "happy" (left), and within-group agreement over all 87 repeated prompts against the permutation null (right).

Table 4.8: Within-group agreement of the image reader across the 87 prompts drawn two or more times (336 records; 43 groups are pairs, whose modal agreement cannot fall below 50%), against a permutation null that shuffles the image readings over all 1,012 analysed records and re-cuts them into the same groups (10,000 draws, seed 7).

Quantity	Observed	Null mean (sd)	z	p
Same family, mean over groups	68.8%	59.4% (2.1%)	+4.5	1.0×10^{-4}
Same valence, mean over groups	74.5%	63.7% (2.1%)	+5.0	1.0×10^{-4}
Fully consistent groups (of 87)	28	14.5 (3.2)	+4.2	2.0×10^{-4}

Table 4.9: Prompt-reader family (rows) against image-reader family (columns) on the 607 emotional records. Recall = row diagonal share, with a clustered-bootstrap 95% CI. Cohen's $\kappa = 0.23$ [0.18, 0.29]. The image reader never returns disgust and returns fear 4 times in 1,012 records.

Prompt \ image	joy	sadness	anger	fear	surprise	disgust	neutral	n	Recall [CI]
joy	261	26	18	1	33	0	63	402	64.9% [59, 70]
sadness	27	40	20	1	5	0	6	99	40.4% [31, 51]
anger	1	2	6	0	1	0	2	12	50.0% [22, 78]
fear	0	6	3	0	3	0	4	16	0.0% [0, 0]
surprise	32	9	3	0	15	0	18	77	19.5% [11, 28]
disgust	0	0	0	0	1	0	0	1	0.0% [0, 0]
image total	321	83	50	2	58	0	93	607	

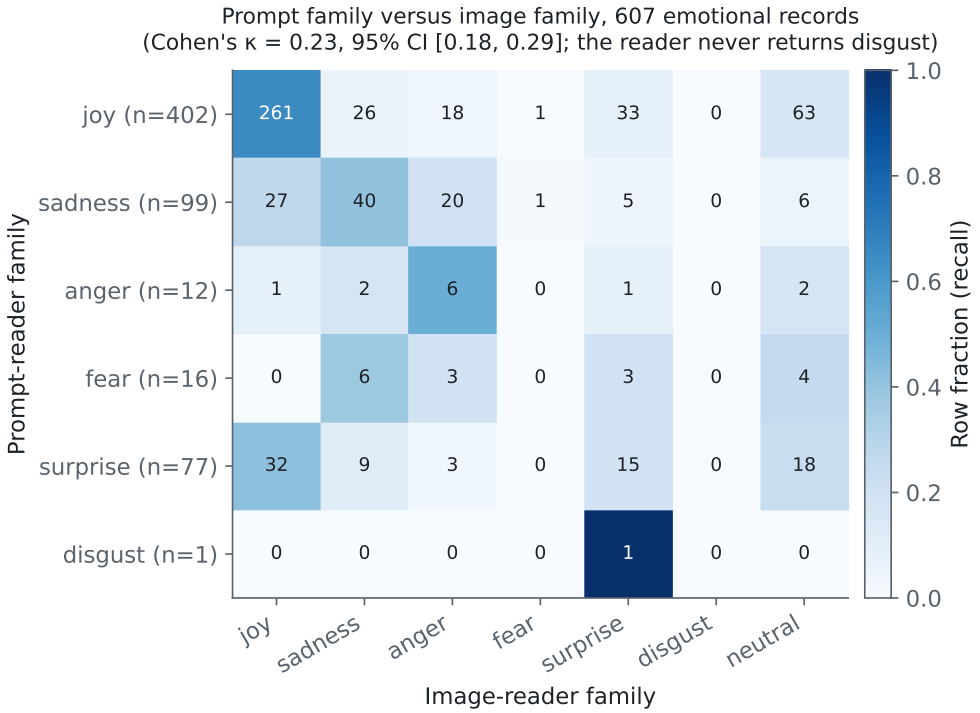


Figure 4.5: Prompt-reader family (rows) against image-reader family (columns) on the 607 emotional records; cell = count, shading = row fraction.

Table 4.10: Cohen's κ for family agreement (7 families), with clustered-bootstrap 95% CIs and the Landis & Koch (1977) band. The neutral-labelled subset is degenerate ($\kappa = 0$ by construction) and is omitted.

Subset	n	κ	95% CI	Band
emotional	607	0.23	[0.18, 0.29]	fair
face	402	0.30	[0.23, 0.37]	fair
scene	205	0.13	[0.06, 0.19]	slight
all analysed	1012	0.21	[0.16, 0.25]	fair
distinct emotional	452	0.23	[0.17, 0.29]	fair
distinct face	319	0.29	[0.21, 0.37]	fair
distinct scene	133	0.09	[0.01, 0.17]	slight

Table 4.11: Top-1 family marginals of the two readers over all 1,012 analysed records.

Family	Prompt reader	Image reader
joy	402	461
sadness	99	141
anger	12	89
fear	16	4
surprise	77	108
disgust	1	0
neutral	405	209

4.3 Robustness to repeated prompts and multiple comparisons

4.4 Specificity check: prompts the text reader labels neutral

4.5 Two image readers, like for like

4.6 Consistency across repeated prompts

4.7 Per-family agreement

4.8 The describer

Table 4.12: The describer's answers on all 1,012 analysed records: people count crossed with whether the facial-expression answer contains an expression word. The face rule is the top-left cell. Of the 421 scene-classified records, 414 report no people and 406 of those still contain an expression word (332 answer smile, smiling or happy).

	Expression word present	Expression word absent
People ≥ 1	591	7
People $= 0$	406	8

4.9 Error taxonomy

Table 4.13: Primary failure mode of the 38 high-confidence disagreements (both readers' top-1 score ≥ 0.8, families differ), assigned from the drawing. Describer faults (a+b) = 19; image reader (c) = 5; prompt side (d+f) = 12; generator (e) = 2.

Mode	Description	n	Exemplar ids
(a)	describer invented a face or expression where none is drawn	9	18, 19, 242
(b)	describer read the expression wrongly	10	39, 192, 202
(c)	reader mis-inferred from a correct description	5	89, 107, 231
(d)	prompt ambiguous or mixed	11	65, 66, 93
(e)	drawing does not depict the prompt	2	137, 771
(f)	prompt reader wrong	1	436

Table 4.14: Description-template ablation: the NLI stage re-run on descriptions rebuilt from the cached describer answers with one clause removed or fixed; the prompt side and the null procedure unchanged. Valence-sign and family agreement per variant on the emotional prompts and their face/scene halves.

Variant	Subset	n	Sign obs	null	z	Family obs	null	z
baseline	emotional	607	64.0%	45.6%	+11.1	53.0%	38.8%	+10.3
	face	402	73.9%	53.5%	+10.4	61.9%	45.8%	+10.3
	scene	205	42.9%	29.6%	+5.8	35.6%	26.2%	+4.4
drop_expression	emotional	607	15.3%	11.4%	+5.1	20.3%	16.6%	+3.8
	face	402	18.9%	14.7%	+3.8	25.9%	20.5%	+3.8
	scene	205	7.6%	4.7%	+2.7	9.3%	6.4%	+2.2
drop_action	emotional	607	61.9%	43.9%	+11.0	50.6%	36.7%	+10.3
	face	402	72.5%	52.2%	+10.4	60.7%	44.5%	+10.4
	scene	205	39.4%	26.8%	+5.6	30.7%	21.9%	+4.6
drop_posture	emotional	607	62.3%	44.5%	+10.7	51.4%	37.9%	+9.7
	face	402	73.1%	53.1%	+10.0	61.4%	45.6%	+10.1
	scene	205	39.4%	26.8%	+5.6	31.7%	24.0%	+3.6
drop_people	emotional	607	68.5%	50.7%	+10.8	57.8%	43.7%	+9.9
	face	402	70.8%	50.7%	+10.4	59.7%	43.6%	+10.3
	scene	205	63.5%	50.9%	+5.0	54.1%	44.5%	+3.8
drop_scene	emotional	607	64.2%	46.0%	+10.7	53.4%	39.2%	+9.8
	face	402	74.4%	53.8%	+10.4	62.4%	46.1%	+10.1
	scene	205	42.4%	29.9%	+5.6	35.6%	26.8%	+3.9
fixed_expression	emotional	607	15.1%	10.8%	+5.9	21.3%	17.5%	+3.9
	face	402	17.5%	12.8%	+4.4	25.1%	19.7%	+4.0
	scene	205	10.0%	6.6%	+2.8	13.7%	11.2%	+1.5
people_aware	emotional	607	51.7%	36.9%	+10.1	47.1%	34.5%	+8.8
	face	402	73.9%	53.5%	+10.4	61.9%	45.8%	+10.3
	scene	205	4.7%	3.4%	+1.6	18.0%	17.3%	+0.6
people_aware_no_expression	emotional	607	55.3%	39.5%	+10.5	48.1%	35.0%	+9.6
	face	402	73.9%	53.5%	+10.4	61.9%	45.8%	+10.3
	scene	205	15.9%	11.6%	+2.7	21.0%	16.8%	+2.2
control_full_template	emotional	607	64.0%	45.6%	+11.1	53.0%	38.8%	+10.3
	face	402	73.9%	53.5%	+10.4	61.9%	45.8%	+10.3
	scene	205	42.9%	29.6%	+5.8	35.6%	26.2%	+4.4

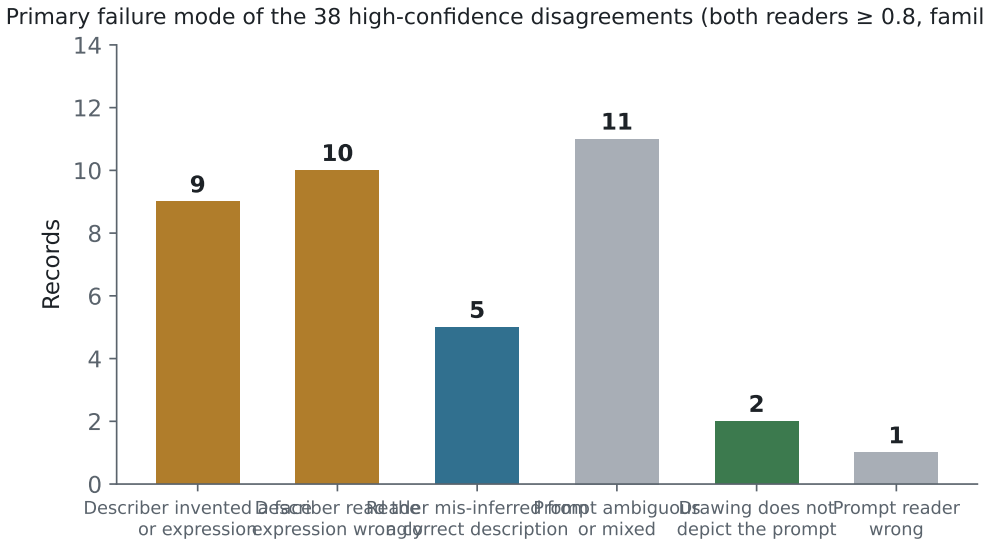


Figure 4.6: Primary failure mode of the 38 high-confidence disagreements, assigned from the drawing.

4.10 Description-template ablation

4.11 Single-coder reliability check

[Human-check table pending: code `outputs/human_check/labels.csv`*, run* `human_check.py score`*, then* `export_tex.py`*.]*

4.12 Summary of findings

Chapter 5

Discussion

5.1 What the results establish

5.2 The describer is the weak link

5.3 Circularity of the face/scene split

5.4 What is not claimed

5.5 Relation to prior work

5.6 Limitations

5.7 Threats to validity

Chapter 6

Conclusions and Further Work

6.1 Conclusions

6.2 Further work

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Appendices

Appendix A

Reproducibility

A.1 Repository layout

- `emoviz/emoviz/metrics.py` — the single definition of every measure, split rule, centroid and the random-pairing null; imported by every script.
- `emoviz/emoviz/data.py` — the loader: filename schemes, timestamps, prompt groups, name redaction at load time.
- `emoviz/preprocess_names.py` — the three-stage redaction procedure; the decision table travels with the dataset.
- `emoviz/describe_extract.py` — the pipeline (BLIP-VQA → template → BART-MNLI); crash-safe, resumable.
- `emoviz/analysis_final.py` — splits, ladder, nulls, distinct-prompt re-run; writes `outputs/analysis_final.json`.
- `emoviz/significance.py` — clustered bootstrap and clustered permutation test of the face/scene difference.
- `emoviz/inference.py` — clustered CIs for every subset, Cohen's κ, Holm–Bonferroni, seed stability.
- `emoviz/method_comparison.py` — like-for-like image-reader comparison with paired tests.
- `emoviz/data_audit_consistency.py` — data accounting and the repeated-prompt consistency with its permutation null.
- `emoviz/audits.py` — control contamination, describer leak and copy rate, per-family recall.
- `emoviz/error_taxonomy.py` — the high-confidence disagreement census and failure modes.
- `emoviz/ablate_nli.py` — description-template ablation (NLI stage only).

- `emoviz/human_check.py` — the single-coder sample, contact sheets and scoring.
- `emoviz/make_figures.py` and `emoviz/export_tex.py` — every figure and every number in this document, generated from the JSON result files.
- `emoviz/app.py` and `emoviz/web/` — the demonstrator.
- `emoviz/test.sh` and `emoviz/test_numeric.py` — the smoke suite and the pinned numeric checks.
- `emoviz/outputs/` — cached model outputs, result JSON, figures, the frozen direct-CLIP baseline.

A.2 Run order

1. `bash emoviz/setup.sh` — creates the environment from the pinned `requirements.txt`.
2. `describe_extract.py` — the pipeline over all images (cached; re-run only to regenerate).
3. `analysis_final.py` — the main analysis (seconds; uses the cached outputs).
4. `significance.py`, `inference.py`, `method_comparison.py`
5. `data_audit_consistency.py`, `audits.py`, `error_taxonomy.py`, `ablate_nli.py`
6. `human_check.py sample`; code `outputs/human_check/labels.csv`; `human_check.py score`
7. `make_figures.py`; `export_tex.py`; then build this document (`pdflatex`, `bibtex`, `pdflatex` twice).
8. `bash emoviz/test.sh` — runs everything above that is fast, plus `test_numeric.py`.

All scripts are run with `emoviz/.venv/bin/python` from the `emoviz/` directory.

A.3 Environment

Appendix B

Instruments

B.1 The five describer questions

The describer is asked, in this order, and its answers are stored per record:

1. *What is the facial expression?* (`expression`)
2. *What is the person doing?* (`action`)
3. *What is the body posture?* (`posture`)
4. *How many people are there?* (`people`)
5. *What is in the picture?* (`scene`)

The word "emotion" appears in none of them.

B.2 The description template

The five answers are composed into one sentence:

```
A drawing of {people} showing {scene}.  The person is {action}, {posture}.
The facial expression is {expression}.
```

The template is applied to every record, including those where `people` is "0" or "none"; the ablation variants remove one clause at a time or fix `expression` to "unclear".

B.3 The image reader's hypothesis

Zero-shot classification by entailment with the hypothesis template

```
This scene conveys {family}.
```

scored for each of the seven families; the three highest entailment scores are kept.

B.4 GoEmotions labels to families

joy admiration, amusement, approval, caring, desire, excitement, gratitude, joy, love, optimism, pride, relief

sadness disappointment, embarrassment, grief, remorse, sadness

anger anger, annoyance, disapproval

fear fear, nervousness

surprise surprise, realization, confusion, curiosity

disgust disgust

neutral neutral

Family scores are the sums of the member probabilities, normalised to a distribution.

B.5 Family centroids on the circumplex

Family	Valence	Arousal
joy	+0.80	0.60
sadness	−0.75	0.30
anger	−0.70	0.80
fear	−0.65	0.75
surprise	+0.30	0.80
disgust	−0.70	0.60
neutral	0.00	0.30

Hand-set from the standard placement of the basic emotions on the circumplex; the minimum of Eq. 3.1 over these points is 0.412.

B.6 Readable expression words

A record is face-classified when the describer reports at least one person and its expression answer contains one of:

> smile, smiling, happy, sad, angry, serious, surprised, surprise, calm, excited, crying, laughing, frowning, scared, shocked, worried, concentration, concentrating, grin, grinning, neutral, confused, determined, pensive, sleepy, sleeping.

The people count is parsed from digits or number words (none, one, two, . . . , several, many, crowd, group, lots); a non-empty answer that is neither is read as one person present.

B.7 Name redaction procedure

1. **Detect.** Candidates from three detectors: a given-name gazetteer (7,944 names), mid-sentence capitalised tokens absent from a 236k-word English dictionary, and naming phrases ("called X", "my friend X"). High recall by design, because most prompts are written entirely in lower case.
2. **Review.** Every candidate has a recorded decision (redact or keep) with a reason and, for redactions, the record it applies to. The audit fails if any candidate is unreviewed or if a redaction entry matches nothing.
3. **Apply.** At load time, per record, replacing the token with `[name]`; consecutive placeholders collapse to one. The files on disk are never modified. Illness, grief and place names are kept.

B.8 Single-coder check: coding instructions

The coder works from numbered contact sheets that show only the drawings, and fills four columns per drawing:

person y / n — is at least one person or face drawn?

face y / n — is there a face whose expression can be read? (n if no person)

valence + / − / 0 — the feeling the drawing gives: positive, negative, neither

family joy / sadness / anger / fear / surprise / disgust / neutral

The sample is fixed by seed (120 drawings: 40 face-classified, 40 scene-classified with no people reported, and every record with a vague or word-only people answer, topped up equally from the first two strata). The prompt, the description and the model readings are not shown during coding.